

REFLECTION ON THE MAGIC OF e^{Xt}

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In general, given variables X and t , if we raise e to the power Xt , differentiate k times with respect to t , then set $t = 0$, we get back X^k , namely,

$$\left. \frac{d^k}{dt^k} e^{Xt} \right|_{t=0} = X^k.$$

This is a consequence of the fact that the Taylor series of e^x is

$$e^x = \sum_{i=0}^{\infty} \frac{x^i}{i!} \implies e^{Xt} = \sum_{i=0}^{\infty} \frac{(Xt)^i}{i!}.$$

Mathematicians are quick to realize that if we just raise e to the power Xt , we effectively have stored all the X^k for $k = 0, 1, 2, \dots$ in one compact expression, e^{Xt} , for later retrieval via differentiation.

WHEN X IS A RANDOM VARIABLE

When X is a random variable, and thanks to the linearity of expectation, we can store all the *moments*, namely, $\mathbf{E}[X^k]$ for $k = 0, 1, 2, \dots$ in the expectation of that compact expression, i.e. $\mathbf{E}[e^{Xt}]$. This, of course, is known as the *Moment Generating Function* (MGF) of X with the usual notation:

$$M_X(t) = \mathbf{E}[e^{Xt}].$$

It is a **moment storage device** — it doesn't generate anything, but one can pull out the k th moment by differentiating k times (and setting $t = 0$), namely,

$$\left. \frac{d^k}{dt^k} M_X(t) \right|_{t=0} = \left. \frac{d^k}{dt^k} \mathbf{E}[e^{Xt}] \right|_{t=0} = \mathbf{E}[X^k].$$

As one can see, this works by construction.

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